DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY
UTTAR PRADESH, LUCKNOW

Evaluation Scheme & Syllabus

For

B.Tech. 3rd Year

(Chemical Engineering)

Based On

AICTE MODEL CURRICULUM

(Effective from the Session: 2024-25)

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH III YEAR V SEMESTER CHEMICAL ENGINEERING

SEMESTER- V													
Sl. No	**Subject Codes**	**Subject**	**Periods**			**Evaluation Scheme**				**End Semester**		**Total**	**Credit**
			L	**T**	**P**	**CT**	**TA**	**Total**	**PS**	**TE**	**PE**		
1	BCH501	Mass Transfer -I	3	1	0	20	10	30		70		100	4
2	BCH502	Chemical Reaction Engineering - II	3	1	0	20	10	30		70		100	4
3	BCH503	Process Dynamics and Control	3	1	0	20	10	30		70		100	4
4	BCH 051-054	Departmental Elective-I	3	0	0	20	10	30		70		100	3
5	BCH 055-058	Departmental Elective-II	3	0	0	20	10	30		70		100	3
6	BCH551	Mass Transfer-I Lab	0	0	2				50		50	100	1
7	BCH552	PDC Lab	0	0	2				50		50	100	1
8	BCH553	Process Modelling and Simulation Lab	0	0	2				50		50	100	1
9	BCH555	Mini Project or Internship Assessment*****	0	0	2				100			100	1
10	BNC501/ BNC502	Constitution of India/ Essence of Indian Traditional Knowledge	2	0	0	20	10	30		70			NC
11		MOOCs (Essential for Hons. Degree)											
		Total	**17**	**3**	**8**							**900**	**22**

*****The Mini Project or internship (4 weeks) conducted during summer break after IV semester and will be assessed during V semester.

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH III YEAR VI SEMESTER CHEMICAL ENGINEERING

SEMESTER- VI													**SESSION 2024-25**
Sl. No	**Subject Codes**	**Subject**	**Periods**			**Evaluation Scheme**				**End Semester**		**Total**	**Credit**
			L	**T**	**P**	**CT**	**TA**	**Total**	**PS**	**TE**	**PE**		
1	BCH601	Mass Transfer -II	3	1	0	20	10	30		70		100	4
2	BCH602	Transport Phenomenon	3	1	0	20	10	30		70		100	4
3	BCH603	Chemical Technology	3	1	0	20	10	30		70		100	4
4	BCH061-064	Departmental Elective-III	3	0	0	20	10	30		70		100	3
5		Open Elective-I [Annexure - B(iv)]	3	0	0	20	10	30		70		100	3
6	BCH651	Chemical Technology Lab	0	0	2				50		50	100	1
7	BCH652	Mass Transfer-II Lab	0	0	2				50		50	100	1
8	BCH653	Technical Presentation	0	0	2				100			100	1
9	BNC601/ BNC602	Constitution of India/ Essence of Indian Traditional Knowledge	2	0	0	20	10	30		70			
10		MOOCs (Essential for Hons. Degree)											
		Total	**17**	**3**	**6**							**800**	**21**

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH III YEAR CHEMICAL ENGINEERING

(DEPARTMENT ELECTIVE SUBJECTS)

DEPARTMENT ELECTIVE – I

BCH-051	Computational Fluid Dynamics
BCH-052	Optimization Techniques
BCH-053	Numerical Methods for Chemical Engineer
BCH-054	Statistical Design of Experiments

DEPARTMENT ELECTIVE –II

BCH-055	Quality Assurance & Control
BCH-056	Process Flow Sheet Simulation
BCH-057	Process Integration
BCH-058	Intellectual Property Rights & Standardization

DEPARTMENT ELECTIVE- III

BCH-061	Fundamentals of Polymer Engineering
BCH-062	Sustainability of Environment
BCH-063	Colloid Surface & Interfacial Phenomena
BCH-064	Environment Impact Assessment

DR. A.P.J ABDULKALAM TECHNICAL UNIVERSITY, LUCKNOW

B.TECH.III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 501	**COURSE TITLE: MASS TRANSFER–I**
EXAM DURATION: 3 HOURS	**SEMESTER: V (ODD)**
L: T: P:: 3 :1 : 0 CREDITS: 4	**PREREQUISITE: CHEMICAL ENGG. FLUID MECHANICS, THERMODYNAMICS**

OBJECTIVES:

1. To provide students with a comprehensive understanding of the fundamental principles of mass transfer and the various mechanisms involved in mass transport processes.
2. To impart knowledge on molecular diffusion, convective mass transfer, and the determination of mass transfer coefficients in various phases.
3. To develop the ability to analyze and design equilibrium-based separation processes such as distillation, absorption, and drying.
4. To introduce students to the design and operational aspects of mass transfer equipment like distillation columns, packed towers, and tray towers.

COURSE OUTCOMES:

On successful completion of the course, the student will be able to:

1. Demonstrate a clear understanding of the fundamental concepts of mass transfer, including diffusion, mass transfer coefficients, and the concept of driving force in mass transport processes.
2. Analyze and solve problems related to mass transfer operations like distillation.
3. Analyze the absorption phenomenon using both theoretical and empirical approaches.
4. Design and optimize simple equilibrium-staged separation processes, such as binary distillation and absorption columns, considering energy efficiency and cost..
5. Calculate mass transfer rates in various systems and determine mass transfer coefficients using appropriate correlations and experimental data.
6. Analyze the drying phenomenon.

REFERENCEBOOKS:

S.NO.	NAMEOFAUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	"Mass Transfer Operations" by Robert E. Treybal, McGraw-Hill Education	2012
2.	McCabe and Smith: Unit Operations in Chemical Engineering II, 7th Edition 7e, McGraw-Hill	2017
3.	Coulson and Richardson: Heat and Mass Transfer: Fundamentals and Applications VolI-B,7e,	2018
4.	Principles of Mass Transfer and Separation Processes by B.K. Dutta,2e, PHI Learning Pvt. Ltd.	2015
5.	Geankoplis,C.J.―TransportProcessesandUnitOperations‖,3e ,PrenticeHall(I),	2003

COURSE DETAILS:

UNITS	CONTENTS	LECTRE HOURS
I	**Introduction to Mass Transfer & Diffusion:** Overview of Mass Transfer Operations, Classification of Mass Transfer Operations, Molecular Diffusion in Fluids: Fick's Law, Diffusivity, and Factors Affecting Diffusion, Steady-State Diffusion in Gases and Liquids, Equimolar Counter Diffusion, Interphase Mass Transfer, Mass Transfer Coefficients, Film Theory, Penetration Theory, Surface Renewal Theory, Two-Film Theory, and Overall Mass Transfer Coefficient	**8**
II	**Absorption and Stripping:** Principles of Gas Absorption, Solubility of Gases in Liquids, Equilibrium in Gas Absorption, Design of Packed Columns: HTU and NTU Concepts, Design of Tray Columns: Plate Efficiency, Murphree Efficiency, Determination of Column Diameter and Column Height (HETP), Absorption with Chemical Reaction: Design Considerations	**8**
III	**Distillation: I-** Fundamentals, Overview and Importance, Types of Distillation: Batch vs. Continuous, Binary vs. Multicomponent, Vapor-Liquid Equilibrium (VLE): Raoult's Law and Dalton's Law, VLE Data and Diagrams: T-x-y, P-x-y, Ideal and Non-Ideal Systems, Azeotropes,, Single Stage Distillation: Differential Distillation, Flash Vaporization, Vacuum Distillation, Molecular Distillation, Steam Distillation	**8**
IV	**Distillation-II:** Multistage and Advanced Distillation Techniques: Continuous Distillation of Binary Mixtures: Multistage Contact Operations, Characteristics of Multistage Towers., McCabe-Thiele Method, Ponchon-Savarit Method, Reflux: Maximum, Minimum, and Optimum Reflux, Tray Columns: Types (Bubble Cap, Sieve, Valve), Tray Efficiency, Determination of Height and Column Diameter, Advanced Distillation Methods: Multistage Batch Distillation, Principles of Azeotropic and Extractive Distillation, Introduction to Multicomponent Distillation Systems	**8**
V	**Drying:** Introduction to Drying: Overview of Drying Operations, Moisture Content and its Measurement, Classification of Drying Operations: Batch and Continuous Drying, Drying Principles and Mechanisms, Drying Rate Curves: Constant Rate and Falling Rate Periods, Factors Affecting Drying Rate, Equilibrium Moisture Content and Hysteresis, Calculation of Drying Time, Types of Dryers and Their Design Tray Dryers, Rotary Dryers, Fluidized Bed Dryers, Spray, Industrial Applications Case Studies	**8**
		40

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

B. TECH III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 502	**COURSE TITLE: CHEMICAL REACTION ENGINEERING – II**
EXAM DURATION: 3 HOURS	**SEMESTER: V (ODD)**
L: T: P:: 3 : 1 : 0 CREDITS: 4	**PRE-REQUISITES: Nil**

OBJECTIVE:

1. To impart the basic concepts of chemical reaction engineering, reactors and contacting pattern
2. To develop understanding about reactor analysis and design for heterogeneous reactions
3. To impart knowledge about the Biochemical reactions.
4. To teach fluid-solid reactions using models like the shrinking core model and determine rate-controlling steps.

COURSE OUTCOME:

After successful completion of the course, the students will be able to:

1. Understand the fundamentals of homogeneous and heterogeneous reactions, including catalyst selection and deactivation.
2. Analyze adsorption mechanisms, catalytic reaction mechanisms, and their impact on chemical equilibrium.
3. Explore solid-catalyzed reactions, focusing on pore diffusion resistance, surface kinetics, and reactor performance equations.
4. Evaluate fluid-solid reactions using models like the shrinking core model and determine rate-controlling steps.
5. Study fluid-fluid reaction kinetics, mass transfer, and reactor design based on contactor types and patterns.
6. Gain insights into biochemical reactions, enzyme kinetics, and the design of biochemical reactors for bioprocessing applications.

CO-PO Mapping (1 to 3 scale)

CO/PO	**PO -1**	**PO -2**	**PO -3**	**PO -4**	**PO -5**	**PO -6**	**PO -7**	**PO -8**	**PO -9**	**PO -10**	**PO -11**	**PO -12**
CO-1	3	2	2	1	1	1	2	1	-	-	-	-
CO-2	3	3	2	1	2	2	2	1	-	-	-	-
CO-3	3	3	2	2	2	1	2	1	-	-	-	-
CO-4	3	3	2	2	1	1	1	1	-	-	-	-
CO-5	3	3	2	2	2	1	2	1	-	-	-	-
CO-6	3	2	2	3	2	2	2	1	-	-	-	-
Avg	**3.0**	**2.7**	**2.0**	**1.8**	**1.7**	**1.3**	**1.8**	**1.0**	**-**	**-**	**-**	**-**

REFERENCE BOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/REPRINT
1	Smith, J, M, "Chemical Engineering Kinetics", 3rd Edition, McGraw-Hill	1990
2	Levenspiel, O., "Chemical Reaction Engineering", 3rd Edition, John Wiley,	1998
3	Fogler H.S., Elements of Chemical Reaction Engineering, 4th edition, Prentice Hall of India	2008
4	Daizo Kunii & Octave Levenspiel, "Fluidization Engineering" 2nd Edition, Elsevier (India Print 2005)	2005
5	Coulson and Richardson's Chemical Engineering Volume 3 - Chemical and Biochemical Reactors and Process Control (3rd Edition)	2006

COURSE DETAILS

UNITS	CONTENTS	LECTUREHOURS
I	Introduction to Homogeneous and Heterogeneous reactions, catalysts and Nature of catalysis, Physical properties of catalysts; Classification, preparation, testing and characterization of solid catalysts, catalyst selection, catalyst promoters and inhibitors, catalyst deactivation (no kinetics). Adsorption, physical adsorptionand chemisorption, adsorption isotherms, mechanisms of catalytic reactions, Shifting of equilibrium in chemical reactions	**9**
II	Solid catalysed reactions, the rate equations for surface kinetics, Reaction and diffusion within porous catalysts, Pore diffusion resistance combined with surface kinetics, effectiveness factor and Thiele modulus, various resistances to transfer of reactants to thecatalyst site, intrinsic and global rate of reaction, kinetic regimes, heat effects during reaction, Performance equations for reactors containing porous catalyst particles, design of solid catalytic reactors.	**9**
III	Fluid-solid reactions, selection of a model, shrinking core model for spherical particles of unchanging size, rate of reaction for shrinking spherical particles, determination of rate controlling step, kinetics and design of reactors.	**8**
IV	Fluid-Fluid Reactions, Rate equation, rate equation for straight mass transfer, kinetic regimes of mass transfer and chemical reaction, rate equation for mass transfer and chemical reactions, fluid-fluid reactor design, deciding the contactor type and contactingpattern.	**8**
V	Introduction to Biochemical reactions: Kinetics of Enzyme Fermentation and Microbial Fermentation, understanding of Biochemical Reactors and study of polymerization reactions.	**6**
	Total	**40**

DR. A.P. J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH. III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 503 COURSE TITLE: Process Dynamics & Control
EXAM DURATION: 03 Hours SEMESTER: V (ODD)
L: T: P :: 3 : 1 : 0 CREDITS: 4 PREREQUISITE: Engg. Maths

Course Objectives:

1. To impart knowledge about basic ideas, challenges, techniques, and applications of process controlfor controlling various processes.
2. To teach the fundamental aspects of process dynamics and control, which includes developing dynamic models of processes, control strategies for linear time-invariant systems and instrumentation aspects.

Course Outcomes

1. On completion of this course, the students will be able to:
2. Demonstrate fundamental understanding of process control.
3. Develop transfer function (input-output) and models for linear dynamic processes.
4. Characterize the dynamics and stability of processes based on mathematical analysis.
5. Develop the mathematical model of various chemical processes.
6. Understand different control modes and their application in controlling various processes and the working of different controllers and valves.

CO-PO Mapping (1 to 3 scale)

C/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	3	3	1	1	1	-	-	-	-	-	2
CO-2	3	3	3	2	1	1	1	1	1	-	-	2
CO-3	2	3	2	1	2	2	2	1	-	-	-	1
CO-4	3	2	3	1	1	1	1	1	-	-	-	3
CO-5	2	3	2	2	2	1	-	1	1	-	-	2
CO-6	2	3	2	1	1	2	1	-	-	-	-	1
Avg	**2.5**	**2.8**	**2.5**	**1.3**	**1.3**	**1.3**	**1.3**	**1.0**	**1.0**	**-**	**-**	**1.8**

Reference Book:

1. Coughnaowr, D. R., “Process Systems Analysis and Control”, McGraw-Hill, Inc 2017.
2. Stephanopolous, G., “Chemical Process Control”, Prentice-Hall, 2008.
3. Seborg, D. E., Edgar, T., and Mellichamp, D. A., “Process Dynamics andControl”, John Wiley and Sons, 2016.
4. Bequette, B. W., “Process Control: Modeling, Design, and Simulation”,Prentice-Hall, Inc, LE.
5. Chidambaram, M., “Computer Control of Processes” Narosa PublishingHouse Pvt. Ltd., Ind, (LE) latest Edition.

Course Details

Unit	Content	Contact Hours
I	Introduction to process control: Needs & control aspects of a chemical plant: stirred tank heater, flow in tank, control of heat exchanger, distillation column and reactor, SISO and MIMO control., Introduction to mathematical modeling: State variables and state equations, deviation variables, Measurement Techniques and Instruments for level, flow, temperature and pressure measurements; dead time, modeling with dead time, multivariable system, linearization of nonlinear systems.	**8**
II	Laplace transform/ Inverse Transform: Laplace transform of step, impulse , pulse ,ramp, trigonometric functions, derivative and integral functions, initial value theorem, Final value theorem, Dirac delta functions, Inverse Laplace transform, Inversion by partial fraction, solution of linear differential equation using Laplace transform, transfer function and input–output model, Poles & zeros of system.	**8**
III	First Order System, physical examples of first order systems, Dynamic behavior of first order system: Time constant, steady state gain, response of the system, first order system in series Dynamic behaviors of second order system: study of interacting & non-interacting responses, inverse response, multi-capacity process, over-damped, critically damped, under-damped responses and their characteristics, overshoot, decay ratio, period of oscillation, rise time, ultimate period, delay time, transportaion lag.	**8**
IV	Components of a Control System (Sensors, Controllers, Actuators), Types of Control Systems: Open Loop vs. Closed Loop, Basic Control System Configuration. Block diagram and Development of block diagram. Feedback control system: Basics of feedback control system, Types (Negative and Positive Feedback systems), Types of controllers: On-Off, P, PI, PID controllers (Principle, Transfer function, Mathematical Representation, Characteristics and example). Block diagram reduction to solve feedback loop, Transmission lines, final control elements, Control valves and their characteristics.	**8**
V	Stability and Performance Analysis: Stability Criteria (Definition of Stability in Control Systems, Characteristic Equation and Stability Analysis), Servo and regulatory problems. Routh-Hurwitz Stability Criterion (Bode Plot: Magnitude and Phase Response, Bode Stability Criteria: Gain Margin and Phase Margin), Nyquist Stability Criterion, Gain Margin and Phase Margin (Definition, Calculation, and Interpretation. Design of controllers: Simple performance criteria, Time Integral performance criteria, Ziegler Nichols tuning technique, Cohen – coon tuning technique. Introduction to advance control strategies: ratio, cascade, feed forward, override and valve	**8**
		40

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

B. TECH. III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 551	**COURSE TITLE: MASS TRANSFER-1 LAB**
EXAM DURATION: 2 HOURS	**SEMESTER: V**
L: T: P :: 0 : 0 : 2 CREDIT: 1	**PRE-REQUISITE: MASS TRANSFER-1**

OBJECTIVE:

1. To impart knowledge about the basic fundamental principles of mass transfer by performing different experiments
2. To make them correlate theory and practical processes by experimentation.

COURSE OUTCOMES:

On successful completion of the course, the student will be able to:

1. Understand the principles of molecular diffusion and basic laws of mass transfer.
2. Utilize mass transfer concepts to design gas absorption systems.
3. Understand the basics of all types of distillation process and its application
4. Apply the concept and mechanism of drying operations.
5. Understand the concept of the crystallization process
6. Calculate reflux ratio, optimum reflux etc.

CO-PO Mapping (1 to 3 scale)

CO/PO	**PO-1**	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**	**PO-7**	**PO-8**	**PO-9**	**PO-10**	**PO-11**	**PO-12**
CO-1	3	3	3	1	1	1	2	1	-	-	-	2
CO-2	2	3	2	1	2	2	2	1	-	-	-	2
CO-3	3	3	3	2	2	1	2	1	-	-	-	2
CO-4	2	3	2	1	1	1	1	1	-	-	-	2
CO-5	3	3	3	1	2	1	2	1	-	-	-	2
CO-6	3	2	3	1	2	2	2	1	-	-	-	2
Avg	**2.7**	**2.8**	**2.7**	**1.2**	**1.7**	**1.3**	**1.8**	**1.0**	**-**	**-**	**-**	**2.0**

REFERENCE BOOKS:

S. NO.	**NAME OF AUTHORS/BOOKS/PUBLISHERS**	**YEAR OF PUBLICATION/REPRINT**
1.	Robert. E. Treybal - Mass Transfer Operations II, 4e, Mc Graw Hill, NY,	2017
2.	McCabe and J.M.Smith- Unit Operations in Chemical Engineering‖ , 7e, , McGraw Hill	2005
3.	Coulson and Richardson -Heat and Mass Transfer: Fundamentals and applications, Vol I-B, 7e,	2021

5.	Geankoplis, C.J. ―Transport Processes and Unit Operations‖, 5e, Prentice Hall (I),	2018

COURSE DETAILS:

S. NO.	LIST OF EXPERIMENT
1	To Determine the Gas-phase mass-transfer coefficient in wetted wall column
2	To Determine the diffusion co-efficient of an organic vapor (naphthalene) in air.
3	To estimate the Solid liquid mass-transfer coefficient for dissolution of benzoic acid in water.
4	To study the absorption of a gas in a packed column and calculation of NTU and HTU
5	To study flooding and loading characteristic of Packed bed absorption column
6	To study working and operation of the cooling tower
7	To study the fluidized bed drying
8	To Study drying characteristics of solids in a batch tray dryer under constant drying conditions
9	To find out crystal yield in batch crystallizer
10	To study crystallization operation using open pan Crystallizer

DR. A.P. J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH. III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 552 COURSE TITLE: PDC Lab
EXAM DURATION: 2 HOURS SEMESTER: V (ODD)
L: T: P :: 0 : 0 : 2 CREDITS: 1 PREREQUISITE: NIL

Course Objectives:

1. To provide knowledge to apply fundamental principles of chemical process dynamics and control.
2. To teach dynamics of typical chemical engineering processes and experiments involving control loop design.
3. To inculcate the hands on practice attitude in students to operate different types of controllers and valves when they work in these fields.

Course Outcomes

1. On completion of this course, the students will be able to:
2. Demonstrate fundamental understanding of process control.
3. Develop transfer function (input-output) and models for linear dynamic processes.
4. Characterize the dynamics and stability of processes based on mathematical analysis.
5. Develop the mathematical model of various chemical processes.
6. Understand different control modes and their application in controlling various processes and the working of different controllers and valves.

CO-PO Mapping (1 to 3 scale)

CO/PO	**PO-1**	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**	**PO-7**	**PO-8**	**PO-9**	**PO-10**	**PO-11**	**PO-12**
CO-1	3	3	3	1	1	1	-	-	-	-	-	2
CO-2	3	3	3	2	1	1	1	1	1	-	-	2
CO-3	2	3	2	1	2	2	2	1	-	-	-	1
CO-4	3	2	3	1	1	1	1	1	-	-	-	3
CO-5	2	3	2	2	2	1	-	1	1	-	-	2
CO-6	2	3	2	1	1	2	1	-	-	-	-	1
Avg	**2.5**	**2.8**	**2.5**	**1.3**	**1.3**	**1.3**	**1.3**	**1.0**	**1.0**	**-**	**-**	**1.8**

Reference Book:

1. Coughnaowr, D. R., “Process Systems Analysis and Control”, McGraw-Hill, Inc 2017.
2. Stephanopolous, G., “Chemical Process Control”, Prentice-Hall, 2008.
3. Seborg, D. E., Edgar, T., and Mellichamp, D. A., “Process Dynamics andControl”, John Wiley and Sons, 2016.
4. Bequette, B. W., “Process Control: Modeling, Design, and Simulation”,Prentice-Hall, Inc, 2003.
5. Chidambaram, M., “Computer Control of Processes” Narosa PublishingHouse Pvt. Ltd., Ind, 1994.

Course Details

S. No.	LIST OF EXPERIMENTS
1	Transient response to single tank system with storage & Flow to (a) step change (b) Impulse change input.
2	Transient response of non-interacting system in series.
3	Transient response of interacting system in series.
4	Study the operation of ON-OFF electronic temperature controller & determination of its performance to control the temperature of a system having capacity to store thermal energy.
5	Study the principle of operation & working of pneumatic servo system with various input functions.
6	Transient response of a CSTR System to step change.
7	Controlling a batch reactor using digital PID controller.
8	Study the dynamics of parallel & counter flow shell & tube heat exchanger.
9	Controlling of Parallel Flow & counter flow STHE using digital PI controller to have desired output.
10	Dynamics characteristics of mercury & water manometers.
11	Study of control value characteristics

DR. A.P. J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

B.TECH. III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 553	**COURSE TITLE: PROCESS MODELING & SIMULATION LAB**
EXAM DURATION: 2 HOURS	**SEMESTER: V (ODD)**
L: T: P :: 0 : 0 : 2 CREDIT: 1	**PREREQUISITE: NIL**

COURSE OBJECTIVES:

1. To impart conceptual knowledge about the modelling & simulation techniques of chemical processes.
2. To teach various simulation approaches and basic knowledge of simulators.
3. To provide knowledge about skills in using process simulators for solving chemical engineering processesproblem.

COURSE OUTCOMES:

On completion of this course, the students will be able to:

1. Understand the fundamentals of process modeling and simulation, including key concepts and techniques.
2. Apply MATLAB to solve process modeling problems and simulate chemical processes.
3. Gain proficiency in using Aspen Plus/CHEMCAD for simulating individual equipment and analyzing VLE data.
4. Design and optimize distillation columns, evaporators, and heat exchangers using simulation tools.
5. Simulate absorption, reaction, and diffusion processes in various reactors using a two-film model.
6. Implement material and energy balances for reactor-separator systems and optimize process performance.

CO-PO Mapping (1 to 3 scale)

CO/PO	**PO -1**	**PO -2**	**PO -3**	**PO -4**	**PO -5**	**PO -6**	**PO -7**	**PO -8**	**PO -9**	**PO -10**	**PO -11**	**PO -12**
CO-1	3	2	2	2	3	2	2	1	1	1	2	3
CO-2	3	3	2	2	3	2	1	1	1	1	2	3
CO-3	3	3	3	3	3	2	2	1	1	2	2	3
CO-4	3	3	3	3	3	2	2	1	1	2	2	3
CO-5	3	3	3	3	3	2	2	1	1	2	2	3
CO-6	3	3	3	3	3	2	2	1	1	2	2	3
Avg	**3.0**	**2.8**	**2.7**	**2.7**	**3.0**	**2.0**	**1.8**	**1.0**	**1.0**	**1.7**	**2.0**	**3.0**

REFERENCE BOOKS:

S. No.	**NAME OF AUTHORS/BOOKS/PUBLISHERS**	**YEAR OF PUBLICATION/ REPRINT**

1.	Luyben W.L., Process Modeling, Simulation and Control for Chemical Engineering, McGraw-Hill.	1998
2.	Babu, B.V., Process Plant Simulation, Oxford University Press.	2004
5.	"Process Modelling and Simulation in Chemical, Biochemical and Environmental Engineering" by Ashok Kumar Verma, 1e, CRC Press	2020
6.	Jana, Amiya K. "Process modelling and control using ASPEN". PHI Learning Pvt. Ltd.	2011
7.	Kamal, I. M. AL-Malah, Aspen plus: chemical engineering applications,2e, Wiley	2022
8.	Juma Haydary, Chemical Process Design and Simulation: Aspen Plus and Aspen HYSYS Applications, Wiley	2019
10.	Yeong Koo Yeo, Chemical Engineering Computation with MATLAB, CRC Press	2018
11.	NPTEL Lectures	

COURSE DETAILS:

S.NO.	LIST OF EXPERIMENTS
1.	Introduction to Process Modeling & Simulation.
2.	Practice examples of process Modeling & simulation and solution of problems using MATLAB
3.	Introduction to Aspen Plus and Simulation of individual equipment using ASPEN Plus.
4.	To calculate the VLE data for ideal mixture and various activity coefficient models by using ASPEN Plus.
5.	To determine the Composition of vapor and liquid streams in a flash distillation still using VLE data.
6.	To apply material balance/enthalpy balance in the plate columns.
7.	To study the absorption, reaction and diffusion processes in a contact reactor/bubble absorber/packed tower/plate column through a two film model.
8.	To simulate liquid –liquid extraction column.
9.	To design and optimize a single effect and multiple effect evaporator.
10.	To design of a shell and tube heat exchanger.
11.	To simulate the CSTR/PFR model and compute the component mole fractions in the product stream.
12.	To simulate the laminar flow of water through a constant diameter circular pipe.
13.	To apply complete material and energy balance for a given reactor-separator system.

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B. TECH. III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 051	**COURSE TITLE: COMPUTATIONAL FLUID DYNAMICS**
EXAM DURATION: 3 HOURS	**SEMESTER: V (ODD)**
L: T: P:: 3 : 0 : 0 CREDITS: 3	**PREREQUISITES: Basic Programming Language**

COURSE OBJECTIVES:

1. To introduce the widely used techniques in the numerical solution of fluid equations.
2. To disseminate the understanding of issues that arise in the solution of such equations, and modern trends in CFD.
3. To emphasize on 'learning by doing'.

COURSE OUTCOMES:

On completion of this course, the students will be able to:

1. Classify the basic equations of fluid dynamics.
2. Understand Basic space and time discretization methods. - Numerical solution of advection, diffusion and stationary problems.
3. Do the numerical solution of Grid Generation, FDM.
4. Analyze the accuracy and stability of finite difference methods for model equations.
5. Work on programming projects.
6. Apply concepts for various problems

CO-PO Mapping (1 to 3 scale)

CO/PO	PO -1	PO -2	PO -3	PO -4	PO -5	PO -6	PO -7	PO -8	PO -9	PO -10	PO -11	PO -12
CO-1	3	2	2	1	2	1	2	-	-	-	-	3
CO-2	2	3	2	1	3	1	2	2	-	-	-	2
CO-3	2	2	2	3	2	2	2	2	-	-	-	2
CO-4	3	3	3	2	1	1	1	1	-	-	-	2
CO-5	2	2	2	1	1	2	2	2	-	-	-	2
CO-6	3	3	3	3	3	2	-	-	-	-	-	2
Avg	**2.5**	**2.5**	**2.3**	**1.8**	**2.0**	**1.5**	**1.8**	**1.8**	**-**	**-**	**-**	**2.2**

REFERENCE BOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	Fletcher C.A.J. "Computational Techniques for Fluid Dynamics, Vol. 1: Fundamental and General Techniques", Springer-Verlag., 2nd edition	1998
2.	Fletcher C.A.J. "Computational Techniques for Fluid Dynamics, Vol . 2: Specific Techniques for Different Flow Categories", Springer-Verlag .	1998

3.	Anderson. J.D., "Computational Fluid Dynamics", McGraw Hill.	2017
4.	Ghoshdastidar P.S., "Computer Simulation of Flow and Heat Transfer", Tata McGraw Hill	1998
5.	Patankar S.V., "Numerical Heat Transfer and Fluid Flow", Taylor and Francis	2017

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	**Basic Concepts of Fluid Flow:** Philosophy of computational fluid dynamics (CFD), review of equations governing fluid flow and heat transfer, simplified flow models such as incompressible, inviscid, potential and creeping flow, flow classification.	**5**
II	**Grid Generation:** Structured and unstructured grids, choice of suitable grid, grid transformation of equations, some modern developments in grid generation in solving the engineering problems.	**3**
III	**Finite Difference Method (FDM)**: Discretization of ODE and PDE, approximation for first, second and mixed derivatives, implementation of boundary conditions, discretization errors, applications to the engineering problems.	**15**
IV	**Finite Element Method (FEM)** Element types: Line, plane, and solid elements, Finite element formulation, Integral formulations for two-dimensional problems, Numerical integration for two-dimensional problems, Interpolation functions: functions for triangular and rectangular elements, Assembly of 2-D finite elements	**9**
V	**Finite Volume Method:** Discretization methods, approximations of surface integrals and volume integrals, interpolation and differential practices, implementation of boundary conditions, application to the engineering problems.	**8**
	Total	**40**

Remark: Basic computational fluid dynamics software recommended for practice of problem

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B.TECH. III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 052	**COURSE TITTLE: OPTIMIZATION TECHNIQUES**
EXAM DURATION: 3 HOURS	**SEMESTER: V (ODD)**
L: T: P:: 3 : 0 : 0 CREDITS: 3	**PREREQUISITE: NIL**

OBJECTIVES:

1. To provide fundamental knowledge to optimize a process plant.
2. To teach the essential features of optimization-related problems.
3. To teach the fundamentals of linear programming and the principle of optimality.

COURSE OUTCOMES:

On completion of this course, the students will be able to:

1. Understand the role of optimization in chemical processing plants.
2. Develop mathematical models for optimizing problems.
3. Analyze the degree of freedom and complexity of optimization solutions.
4. Analyze different methods for unconstrained one-dimensional data search
5. Solve constraint and non-constraint problems
6. Understand algorithms and programming language

CO-PO Mapping (1 to 3 scale)

CO/PO	**PO-1**	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**	**PO-7**	**PO-8**	**PO-9**	**PO-10**	**PO-11**	**PO-12**
CO-1	3	3	2	2	2	2	2	-	-	-	-	3
CO-2	3	3	2	1	3	1	2	2	-	-	-	2
CO-3	2	2	2	3	2	2	2	2	-	-	-	3
CO-4	3	3	2	2	1	2	1	1	-	-	-	2
CO-5	2	2	3	2	1	2	2	2	-	-	-	3
CO-6	3	3	3	3	3	2	1	1	-	-	-	2
Avg	**2.7**	**2.7**	**2.3**	**2.2**	**2.0**	**1.8**	**1.7**	**1.6**	**-**	**-**	**-**	**2.5**

REFERENCE BOOKS:

S. NO.	**Name of Authors/Books/Publishers**	**Year of Publication/ Reprint**
1.	T.F. Edgar and D.M. Himmelblau "Optimization of Chemical Process", McGraw Hill.	2001
2.	K. Urbanier and C. Mc Dermott "Optimal Design of Process Equipment", John Wiley.	2020
3.	Suman Dutta "Optimization Technique in Chemical Process", Cambridge University Press.	2020
4.	Chander Mohan and Kusum Deep "Optimization Technique", New Age Science.	2019
5.	S.S. Rao "Engineering Optimization", Wiley.	2019
6.	A. Ravindran, K M Ragsdell, and G V Reklaitis "Engineering Optimization: Methods And Applications", Jhon Wiley.	2019

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	**Introduction**: Process optimization, Formulation of various process optimization problems and their classification, Basic concepts of optimization-convex and concave functions, Necessary and sufficient conditions for stationary points.	**8**
II	**Optimization of One-Dimensional Functions**: Unconstrained multivariable optimization direct search methods, Bracketing methods: Exhaustive search, Bounding phase, Region elimination methods-Interval halving, Fibonacci search, Golden section search, Point Estimation, Successive quadratic estimation methods.	**8**
III	**Indirect First Order and Second Order Methods**: Gradient-based methods-Newton Raphson, Bisection, Secant, Cubic spline, Root-finding using optimization Techniques.	**8**
IV	**Multivariable Optimization Algorithms**: Optimality criteria, Unidirectional search, Direct search Methods- Evolutionary optimization, Simplex search, Powell's conjugate direction, Gradient-based methods- Cauchy's (steepest descent) method, Newton's method.	**8**
V	**Constrained Optimization Algorithms**: Kuhn-Tucker conditions, Transformation methods, Penalty function method, Method of multipliers, Sensitivity analysis, Direct search for constraint Minimization-Variable elimination method Advanced techniques for optimization	**8**
		40

DR. A.P. J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

B.TECH. III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 053	**COURSE TITTLE: NUMERICAL METHODS FOR CHEMICAL ENGINEER**
EXAM DURATION: 3 HOURS	**SEMESTER: V (ODD)**
L: T: P :: 3 : 0 : 0 CREDITS:3	**PREREQUISITE: NIL**

OBJECTIVES:

1. To impart knowledge about the different types of equations in chemical engineering problems.
2. To teach the solution strategy of differential-algebraic, ordinary, differential, and partial differential equations.

COURSE OUTCOME:

On completion of this course, the students will be able to:

1. Solve the equations with first order and first degree with linear coefficients.
2. Understand and solve unsteady state problems.
3. Provide the solution of various types of equations.
4. Solve linear and non-linear algebraic equations using numerical method
5. To curve fitting using numerical method
6. Apply the above-mentioned strategies solving Chemical engineering problems.

CO-PO Mapping (1 to 3 scale)

CO/PO	**PO-1**	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**	**PO-7**	**PO-8**	**PO-9**	**PO-10**	**PO-11**	**PO-12**
CO-1	3	3	3	2	2	2	-	-	-	-	-	1
CO-2	2	2	3	1	3	2	2	2	-	-	-	2
CO-3	2	2	3	2	2	2	2	1	-	-	-	3
CO-4	3	2	2	2	3	2	1	1	-	-	-	2
CO-5	3	2	3	1	3	2	1	1	-	-	-	2
CO-6	2	2	3	3	3	2	1	1	-	-	-	2
Avg	**2.5**	**2.2**	**2.8**	**1.8**	**2.7**	**2.0**	**1.4**	**1.2**	**-**	**-**	**-**	**2.0**

REFERENCE BOOKS:

S. NO.	**Name of Authors/Books/Publishers**	**Year of Publication/Reprint**
1.	S. K. Gupta, "Numerical Methods for Engineers", New Academic Science Ltd, 3rd Edition	2021
2.	S.C. Chapra and R.P. Canale, "Numerical Methods for Engineers", McGraw Hill, 7th Edition	2023
3.	S.S. Shastry, "Introductory Methods of Numerical Analysis", Prentice Hall of India, 5th Edition	2012
4.	B.S. Grewal, "Numerical Methods in Engineering & Science", Khanna Publishing house.	2017
5.	M.K. Jain, S.R.K. Iyengar and R.K. Jain, "Numerical Methods for Scientific and Engineering Computation", Wiley Eastern.	2021

6.	Pradeep Ahuja, " Introduction to Numerical Method in Chemical Engineering", PHI Learning Pvt Ltd.	2020
7.	Kenneth J. Beers "Numerical Method for Chemical Engineer: Application in MATLAB", Cambridge University Press.	2022
8.	S Elnashaie and F Uligh "Numerical Techniques for Chemical and Biochemical Engineers using MATLAB", Springer.	2018

UNITS	CONTENTS	LECTURE HOURS
I	**Approximations and Errors:** Types of Errors, Significant figures, Accuracy of Numbers, Precision, Error Propagation, Bisection, Secant, Method of False Position or Regula Falsi etc., Convergence of Iterative Methods, Newton Raphson Method, Newton- Raphson Method for Non Linear Equations in Two Variables, Algorithms & Computer Programming for all these Methods in Applications of Chemical Engineering.	**8**
II	**Solution of Linear Equations:** Mathematical Background, Matrix inversion, Gauss Elimination, Gauss Jordan Method, Gauss-Seidel Iteration Method, Jacobi's Method, Gauss Seidel Method, Eigen Value Problem, Algorithms & Computer Programming for all these Methods in Applications of Chemical Engineering.	**8**
III	**Curve Fitting Methods**: of Least Squares, Fitting a Straight Line and a Polynomial, Fitting a Non-linear Function, Fitting Geometric and Exponential Curves, Fitting a Hyperbola, a Trigonometric Function, etc., Algorithms & Computer Programming of Curve Fitting Methods.	**8**
IV	**Numerical Differentiation & Integration:** Differentiation Formula based on Tabulator at Equal and Unequal Intervals, Newton-Cotes Integration Formulas, Trapezoidal Rule and Simpson's $1/3^{rd}$ and $3/8^{th}$ Rule, Algorithms & Computer Programming for all these Methods in Applications of Chemical Engineering.	**8**
V	**Ordinary Differential Equations:** Taylor's Series and Euler's Method, Modifications and Improvements in Euler's Method, Runge-Kutta 2nd Order & 4th Order Methods, Milne's Predictor-Corrector Methods, Boundary Value Problems, Algorithms & Computer Programming for all these Methods in Applications of Chemical Engineering initial value and boundary value problems	**8**
		40

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY
B.TECH III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 054	**COURSE TITLE: STATISTICAL DESIGN OF EXPERIMENTS**
EXAM DURATION: 3 HOURS	**SEMESTER: V (ODD)**
L : T : P :: 3 : 0 : 0 CREDITS: 3	**PRE-REQUISITES:** NIL

OBJECTIVES:

1. To provide advanced concepts of statistical modeling
2. To introduce and emphasize experimental designs for scientific investigations.

COURSE OUTCOMES:

Upon completion of this course, the students will be able to:

1. Familiarize with basic concepts such as random errors, random variables, random sampling, and hypothesis testing
2. Distinguish between determinate and indeterminate errors and quantify them
3. Compare variability due to random errors with variability from controlled process factors
4. Interpret ANOVA results and identify significant factors that influence the experiments
5. Fit empirical models to experimental data using linear regression concepts
6. Optimize processes using response surface methodology

CO-PO Mapping (1 to 3 scale)

CO/PO	**PO-1**	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**	**PO-7**	**PO-8**	**PO-9**	**PO-10**	**PO-11**	**PO-12**
CO-1	3	3	3	1	1	1	-	-	-	-	-	1
CO-2	3	3	3	1	1	1	1	2	-	-	-	2
CO-3	3	3	3	2	1	1	2	2	-	-	-	3
CO-4	3	2	2	1	2	2	1	2	-	-	-	2
CO-5	3	3	3	1	2	2	1	2	-	-	-	1
CO-6	2	2	3	3	3	2	1	1	-	-	-	1
Avg	**2.8**	**2.7**	**2.8**	**1.5**	**1.7**	**1.5**	**1.2**	**1.8**	**-**	**-**	**-**	**1.7**

REFERENCE BOOKS

S.NO	**NAME OF AUTHORS/BOOKS /PUBLISHERS**	**YEAR OF PUBLICATION/ REPRINT**
1.	Montgomery, D. C., G.C. Runger, Applied Statistics and Probability for Engineers. 7th ed. New Delhi: Wiley-India. .	2022
2.	Montgomery, D. C., Design and Analysis of Experiments. 9th ed. New Delhi: Wiley-India, 2011	2017

3.	Myers, R. H., D. C. Montgomery and C. M. Anderson-Cook, Response Surface Methodology. 4the. New Jersey: Wiley.	2022

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	**Introduction:** Overview of the subject, random errors, random variables, random sampling, determinate and indeterminate errors and their analyses.	**5**
II	**Design of experiments:** Systematic planning of experiments, methods for analysis of experimental results. General linear model ANOVA, and model checking **Factorial Designs:** The 2k Factorial Design, blocking and confounding in the 2k factorial design, introduction to two and three-level fractional factorial design.	**11**
III	**Fitting regression Models:** linear and nonlinear regression, general Linear Models and methods for fitting, validating and testing **Response surface methods:** Introduction to response surface methods and experiments with random factors Strategies for experimental design with examples	**6**
IV	**Probability distribution:** Presentation of experimental data, discrete, continuous and normal probability density functions, standard probability distribution functions: Normal, Student's T, chi-square and F distributions	**8**
V	Introduction to Response Surface Methodology, the Method of Steepest Ascent, Experimental Designs for Fitting Response Surfaces- Designs for Fitting the First-Order Model, Designs for Fitting the Second-Order Model, Evolutionary Operation. Advanced topics in design of experiments	**10**
		40

DR. A.P. J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH. III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH-055 COURSE TITLE: Quality Assurance & Control
EXAM DURATION: 3 HOURS SEMESTER: V (ODD)
L: T: P :: 3 : 0 : 0 CREDITS: 3 PREREQUISITE: NIL

Course Objectives:

1. To impart knowledge about quality control and quality assurance in chemical industries.
2. To teach control charts and total quality management.
3. To provide conceptual knowledge of aspects like QC tests, documentation, quality certifications,ISO and SQC

Course Outcomes :

On completion of this course, the students will be able to:

1. Appreciate the importance of quality assurance and control in the chemical industry.
2. Understand the role of ISO for process plants.
3. Learn the manufacturing operations and controls of process plants.
4. Understand the importance of documentation and the scope of quality certifications applicable toindustries.
5. Understand the responsibilities of QA & QC departments.

CO-PO Mapping (1 to 3 scale)

CO/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	3	3	2	2	1	-	1	-	-	-	2
CO-2	2	3	3	2	1	-	-	-	-	-	-	1
CO-3	3	2	2	2	1	1	-	-	-	-	-	2
CO-4	2	3	2	3	-	1	1	-	-	-	-	1
CO-5	2	1	2	3	2	-	1	1	-	-	-	1
CO-6	3	2	3	3	1	-	1	-	-	-	-	1
Avg	**2.5**	**2.3**	**2.5**	**2.5**	**1.4**	**1.0**	**1.0**	**1.0**	-	-	-	**1.3**

Reference Books:

1. Weinberg S., Good Laboratory Practice Regulations, Vol. 69, 2 e , MarcelDekker Series, 2021
2. ICH guidelines (on ICH website), 2016.
3. ISO 9000 and Total Quality Management, 2015.
4. Piotr Konieczka and Jacek Namiesnik “Quality Assurance and Quality Control in the Analytical Chemical Laboratory”, 2e , CRC Press, 2018
5. P.L. Jain “ Quality Control and Total quality Management”, McGraw Hill, 2e, 2018.
6. Ram Babu Sao “Perfect: Quality Assurance and Quality Control”, 2e, Create Space Independent Publishing Platform, 2022.
7. Amitava Mitra “ Fundamentals of Quality Control and Improvement”, 3e, Wiley, 2022.

8. Quality Assurance for the Chemical and Process Industries: A Manual of Good Practices. ASQ Quality Press, 1999.

Course Details

Unit	Content	Contact Hours
I	**Quality:** Definition, Approaches of Quality Management, Inspection & Test, Elements of Total Quality Management (TQM), Deming's approach. **Quality Circle:** Characteristics of Quality Circle, Introduction to Six Sigma and Taguchi concepts.	6
II	**Quality Assurance (QA):** Management principlesin QA, Forms of QA, Quality in material management, Vendor selection & development. **ISO:** Introduction, ISO: 9000/ ISO: 9001/ISO:17025, Registration process, Benefits of ISO.	6
III	**Statistical Quality Control:** SQC tools, Benefits of SQC, Concept of variation, Frequency distribution curve & its types, Normal Distribution curve. **Control chart for variable:** Definition, Formulae & its problems. Control chart patterns and process capability. Problems on X & R chart and Process capability.	6
IV	**Quality Improvement Programme:** Histogram, Charts, Pareto analysis, types of audit, inspection methods. **Sampling:** Definition, types of sampling, importance, benefits and limitations of sampling.	6
V	**Manufacturing operations and controls:** Sanitation of manufacturing premises, packaging operations, the release of the finished product, time limitations on production, expiry date calculation, production record review, handling of waste, and scrap disposal. Case Studies of Process Industries **BIS Standards:** Key points of quality assurance and quality control in BIS standards	6

DR. A.P. J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH. III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH-056 **COURSE TITLE:** Process Flow Sheet Simulation
EXAM DURATION: 3 HOURS **SEMESTER:** V (ODD)
L: T: P :: 3 : 0 : 0 **CREDITS:** 3 **PREREQUISITE:** Numerical methods/Modeling Simulation

Course Objectives:

1. To introduce about flowsheet simulation.
2. To teach solution strategy of steady state and unsteady state systems.

Course Outcomes:

On completion of this course, the students will be able to:

1. Synthesize a flowsheet for the process on paper and implement this in a simulation program.
2. Deal with ASPEN PLUS/HYSIS/PRO II/Design II/UniSim/OLI Pro/Aspen CustomModeler/TK-Solver.
3. Use process flowsheet simulations to solve problems in the chemical industry.
4. Estimate the thermo-physical properties for the chemical species
5. Identify the correct application of model.
6. Design a distillation column, feed height and number of trays in a column.

CO-PO Mapping (1 to 3 scale)

CO/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	3	3	2	1	1	-	-	-	-	-	3
CO-2	3	3	2	2	1	1	1	-	-	-	-	3
CO-3	3	3	3	1	2	1	1	1	-	-	-	3
CO-4	2	2	2	1	2	2	-	2	-	-	-	2
CO-5	2	2	2	2	2	2	-	1	-	-	-	2
CO-6	3	2	2	1	1	1	1	1	-	-	-	2
Avg	**2.7**	**2.5**	**2.3**	**1.5**	**1.5**	**1.3**	**1.0**	**1.3**	**-**	**-**	**-**	**2.5**

Reference Book:

1. Manuals of ASPEN Plus and ChEMCAD
2. Dimian A. C., "Integrated Design and Simulation of Chemical Processes", Elsevier, 2003.
3. Westerberg, A. W., Hutchison, H. P., Motard, R. L. & Winter, P., "Process Flowsheeting", Cambridge University Press, 1979.
4. K.M. Hangos and I. T. Cameron, "Process Modelling and Model Analysis", Academic Press, 2001.
5. A.K. Jana "Process Simulation and Control using ASPEN", PHI Learning Pvt Ltd., 2012.
6. Sakari Kaijaluoto "Process Optimization by Flowsheet Simulation", Technical Research Centre of Finland, 1984.

7. A.W. Westerberg, et al, “Process Flow Sheeting”, Cambridge UniversityPress, 1990.
8. W. F. Ramirez, “Computational Methods for Process Simulation”, 2nd ed.,Butterworths, 1997
9. Kumar, A., “Chemical Process Synthesis and Engineering Design”, Tata McGraw Hill, 1981

Course Details

Unit	Content	Contact Hours
I	**Introduction to Process Simulation:** Different approaches toprocess simulation; modules and components in a process simulationpackage, selection of flowsheet and simulation software. **Process Flow sheeting:** Approaches to flowsheeting, computer aided flow-sheeting, partitioning and tearing a flowsheet.	6
II	**Formalizing the modeling process:** Methods of structuring complex chemical processes, procedures for process modeling; degrees of freedom in a flow sheet. Numerical properties of the model equations. **Numerical methods for steady-state and dynamic systems:** Differential dgebraic equations; Synthesis of reaction systems and synthesis of azeotropic separation systems.	6
III	**Processing Simulation with software:** ASPEN PLUS/HYSIS/PRO II/Design II/ UniSim/ OLI Pro/ Aspen Custom Modeler/ TK-Solver: Features of simulation packages;Graphical User Interface; Example-1: Flashing of Light Hydrocarbons; Example-2: Vinyl chloride monomer (VCM) flow sheet.	6
IV	**Flow sheet Calculations and Model Analysis Tools:** Sensitivity and case-study runs; Design specifications and calculator blocks; Example: sour water systems (CO_2 and H_2S removal for example)	6
V	**Physical Properties:** Overview of physical property system; Property model specifications; Property data requirements and input; Physical property analysis; Introduction to ICARUS (an economic evaluation package inside ASPEN PLUS), Flow sheet Convergence.	6
		30

DR. A.P. J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

B.TECH. III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH057	**COURSE TITLE: PROCESS INTEGRATION**
EXAMDURATION:3 HOURS	**SEMESTER: V (ODD)**
L: T: P::3 :0 : 0 CREDITS:3	**PREREQUISITES: Thermodynamics, HTO**

OBJECTIVES:

1. To disseminate the understanding of the mass targets in the design of processes.
2. To disseminate the understanding of the energy targets in the design of processes.
3. To teach the basics of process integration and its application.
4. To teach the basics of applications of process integration.
5. To impart the pinch analysis, heat exchange networking and industrial applications.

COURSE OUTCOMES:

Upon completion of this course, the students will be able to:

1. Understand the fundamentals of process integration.
2. Perform pinch analysis.
3. Analyze and design heat exchanger networks.
4. Minimize water consumption and waste generation.
5. Do the heat exchange networking
6. Application of networking for industrial problems

CO-PO Mapping (1 to 3 scale)

CO/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO -10	PO -11	PO -12
CO-1	3	3	3	2	-	-	-	-	-	-	-	1
CO-2	3	3	2	1	1	1	1	1	-	-	-	2
CO-3	3	3	2	1	2	1	1	1	-	-	-	3
CO-4	3	3	3	1	3	-	1	2	-	-	-	2
CO-5	2	2	3	1	1	2	1	1	-	-	-	2
CO-6	3	2	2	1	1	1	1	1	-	-	-	2
Avg	**2.8**	**2.7**	**2.5**	**1.2**	**1.6**	**1.3**	**1.0**	**1.2**	**-**	**-**	**-**	**2.0**

REFERENCEBOOKS

S.NO	**NAMEOFAUTHORS/BOOKS/PUBLISHERS**	**YEAR OF PUBLICATION/ REPRINT**
1.	Kemp I. C., "Pinch Analysis and Process Integration: A user Guide on Process Integration for the Efficient Use of Energy", 4th Ed., Butterworth-Heinemann.	2022
2.	Smith R, "Chemical Process Design and Integration",2nd Ed., Wiley.	2020
3.	Shenoy U. V., "Heat Exchanger Network Synthesis",2e, Gulf Publishing Company.	2006

4.	El Halwagi M. M., "Process Integration",7th Ed., Academic Press.	2022

Course details

UNITS	CONTENTS	LECTURE HOURS
I	**Introduction to Process Integration:** Overview of process integration, Historical development, and significance, Understand the fundamental concepts and evolution of process integration	**8**
II	**Pinch Technology:** Basic concept, comparison with energy auditing, the role of thermodynamic laws, the problem addressed by pinch technology. **Key Steps of Pinch Technology:** Data extraction, targeting, designing, optimization, and supertargeting.	**8**
III	**Basic Elements of Pinch Technology:** Grid diagram, composite curve, problem table algorithm, grand composite curve. **Targeting of Heat Exchanger Network (HEN):** Energy targeting, area targeting, number of units targeting, shell targeting, cost targeting.	**8**
IV	**Designing of HEN:** Pinch design methods, heuristic rules, stream splitting, design of maximum energy recovery (MER), design of multiple utilities and pinches, design for threshold problem, loops and paths.	**8**
V	**Heat Integration of Equipment:** Heat engine, heat pump, distillation column, reactor, evaporator, drier, refrigeration systems. **Heat and Power Integration:** Co-generation, steam turbine, gas turbine. **Case Studies in Process Integration**: Industrial applications, Success stories and lessons learned, Analyse real-world applications of process integration in industry	**8**
		40

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

B.TECH III YEAR V SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 058	**COURSE TITLE: Intellectual Property Rights &Standardization**
EXAM DURATION:3 HOURS	**SEMESTER: V(ODD)**
L: T: P:: 3 :0 : 0 CREDITS:3	**PRE-REQUISITE: NIL**

OBJECTIVES:

1. To introduce fundamental aspects of Intellectual Property Rights to students who are going to play a major role in development and management of innovative projects in industries.
2. To disseminate knowledge on patents, patent regime in India and abroad, and registration aspects
3. To disseminate knowledge on copyrights and trademarks & registration
4. To aware about current trends in IPR and Government steps in fostering IPR

COURSE OUTCOMES:

Upon completion of this course, the students will be able to:

1. Adequate knowledge on patent and copyright for their innovative research works
2. During their research career, information in patent documents provide useful insight on novelty of their idea from state-of-the art search.
3. Developing their idea or innovations
4. Pave the way for the students to catchup Intellectual Property as a career-option
5. Gives awareness of international standards to students
6. Understand Bharat Stage emission and standards (BS)

CO/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	2	3	2	2	1	1	-	-	-	-	-	2
CO-2	3	3	2	2	1	1	1	-	-	-	-	2
CO-3	3	2	3	1	1	1	1	1	-	-	-	1
CO-4	3	2	3	1	1	1	-	1	-	-	-	1
CO-5	2	3	3	1	1	-	-	-	-	-	-	1
CO-6	3	2	2	1	1	-	-	-	-	-	-	2
Avg	**2.7**	**2.5**	**2.5**	**1.3**	**1.0**	**1.0**	**1.0**	**1.0**	**-**	**-**	**-**	**1.5**

REFERENCEBOOKS:

S.NO.	NAME OF AUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	Nithyananda, K V. "Intellectual Property Rights: Protection and Management"India, Cengage Learning India Private Limited.	2019
2.	Neeraj, P., & Khusdeep, D. Intellectual Property Rights. India, PHI learning Private Limited.	2014
3.	Ahuja, V K.. Lawrelating to Intellectual Property Rights. India, IN: LexisNexis.	2017
4.	Yadav P. N. "Handbook of Intellectual Property" IN: Bharti Law House	2019
5.	Joshi P.R. "Standardization and Intellectual Property Rights: A Critical Review" IN: Springer	2022

E-resources:

• Subramanian, N., & Sundararaman, M. (2018). Intellectual Property Rights – An Overview. Retrieved from http://www.bdu.ac.in/cells/ipr/docs/ipr-eng-ebook.pdf

• World Intellectual Property Organization. (2004). WIPO Intellectual property Handbook. Retrieved from https://www.wipo.int/edocs/pubdocs/en/intproperty/489/wipo_pub_489.pdf

Reference Journal:

• Journal of Intellectual Property Rights (JIPR): NISCAIR

Useful Websites:

• Cell for IPR Promotion and Management(http://cipam.gov.in/)

• World Intellectual Property Organization (https://www.wipo.int/about-ip/en/)

• Office of the Controller General of Patents, Designs &Trademarks(http://www.ipindia.nic.in/)

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B.TECH III YEAR V SEMESTER CHEMICAL ENGINEERING

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	**Overview of Intellectual Property:** Definition, Nature, and Significance of IPR, Types of Intellectual Property: Patents, Trademarks, Quality marks / Certification marks, Copyrights, Trade Secrets, Geographical Indications, Plant Variety Protection, Protection Mechanism of Intellectual Property and Challenges in IP protection	6
II	**Patents:** Introduction, Types, Patentability Criteria, Filing Process, patents in R&D, protection of chemical pharmaceutical and biotechnological inventions, Patent Rights and Infringement, Concept of quality mark and standardization, Development in quality mark, Quality Council of India and its role, National accreditation body NABCB (National accreditation board for certification bodies), benefits of accreditation, Important legislations: National and International	10
III	**Copyrights:** Introduction, Types of work protected, Ownership and duration, Rights conferred by copyright, digital copyright. Bureau of Indian Standards (BIS) and its role. **Bharat stage emissions:** Standard and Toxicity level with emission limits, technological advances for BS VI compliances, Geo Graphical Indication: Introduction, Purpose and Objectives, Key features, Legislations (National and International), World Trade Organization, Trade-Related Aspects of Intellectual Property Rights(TRIPS),	8
IV	**Trademarks:** Introduction to trademarks, Types of trademarks, Registration Processes of Trademarks, Certificate Marks(Like AG mark and ISI, Hallmark ECO mark, FPO mark etc), Vegetarian and Non Vegetarian Marks. Legislations (National and International like Indian trademark Act 1999) **Plant Variety Protection:** Introduction, Purpose and objective, Key Features, Legislations (National like Protection of Plant Varieties and Farmers' Rights Act, 2001 and International)	6
V	**Other forms of IP Design:** Management of intellectual property right,quality management systems(QMS), ISO-9000for manufacturing, ISO-14000 for environment, ISO-5000 for energy management systems, ISO-22000 for Food safety management systems(FSMS), Information security management system (ISMS), Cyber Law and Digital Content Protection-Unfair Competition-Meaning and Relationship between Unfair Competition IP Laws	10
		40

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B.TECH. III YEAR VI SEMESTER CHEMICAL ENGINEERING

SUBJECTCODE: BCH 601	**COURSE TITLE: MASSTRANSFER–II**
EXAM DURATION: 3 HOURS	**SEMESTER: VI (EVEN)**
L: T: P:: 3 :1 : 0 CREDITS:4	**PREREQUISITE: MASS TRANSFER -I**

OBJECTIVES:

1. To provide students with a comprehensive understanding of the principles and applications of liquid-liquid extraction and leaching processes in separating components from mixtures.
2. To introduce students to the fundamentals of adsorption, including types of adsorbents, adsorption isotherms, and industrial applications of adsorption in gas and liquid phase separations.
3. To impart knowledge on humidification, and types of various humidifiers and dehumidifiers.
4. To familiarize students with the concepts of crystallization, crystal growth, nucleation, and the design and operation of crystallizers for industrial applications.
5. To teach students the design, analysis, and optimization of mass transfer equipment for liquid-liquid extraction, leaching, adsorption, humidification, and crystallization.

COURSE OUTCOMES:

Students completing the course will be able to:

1. Explain the principles of liquid-liquid extraction, including phase equilibrium, distribution coefficient, selectivity, and choice of solvents, and apply them to solve separation problems.
2. Analyze and design equipment for liquid-liquid extraction (mixer-settlers, packed and spray columns) and leaching (agitated vessels, percolation tanks) based on operating parameters, efficiency, and process requirements.
3. Understand adsorption theories, including Freundlich and Langmuir isotherms, and design adsorption systems for gas and liquid phase separations, considering adsorbent selection, breakthrough curves, and regeneration techniques.
4. Analyze humidification and dehumidification processes.
5. Explain the principles of crystallization, including nucleation, crystal growth, and supersaturation, and design crystallization equipment such as batch and continuous crystallizers for separation and purification processes.
6. Evaluate and compare the performance of various mass transfer equipment used in liquid-liquid extraction, leaching, adsorption, humidification, and crystallization processes.

REFERENCEBOOKS:

S.NO.	**NAME OF AUTHORS /BOOKS/ PUBLISHERS**	**YEAR OF PUBLICATION/REPRINT**
1.	"Mass Transfer Operations" by Robert E. Treybal, McGraw-Hill Education	2012
2.	McCabe and J. M. Smith - Unit Operations in Chemical Engineering‖ ,8e, McGraw Hill	2024
3.	Coulson and Richardson- Heat and Mass Transfer: Fundamentals and Applications‖, VolI-B,8e,	2019

4.	"Principles of Mass Transfer and Separation Processes" 3e, by B.K. Dutta, PHI Learning Pvt. Ltd.	2018
5.	Geankoplis,C.J.—TransportProcessesandUnitOperations‖,4e,PrenticeHall(I),	2018

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	**Liquid-Liquid Extraction**: Principles of extraction, ternary liquid equilibria, effect of pressure and temperature, triangular graphical representation concept of theoretical or ideal stage, single stage operation, multistage continuous operation: co-current and cross current operations, stage calculation, Analytical and graphical solutions of single and multistage operations for miscible and immiscible systems, equipment: mixer settlers, plate column, spray and packed column, etc	**8**
II	**Solid /Liquid Extraction (Leaching):** Theory, mechanism, types of leaching, solid - liquid equilibria, constant and variable underflow, batch and continuous operations: single and multistage cross current and counter-current operations, Concep to fan ideal stage, Overall stage efficiency, Determination of number of stages. equipment: percolation tank, agitated vessel, thinkers, classifiers etc	**8**
III	**Humidification and Dehumidification:** Basic Concepts of Humidification and Dehumidification, Importance in Chemical Engineering Processes, Definitions and Terminology: Humidity, Relative Humidity, Dew Point, Wet Bulb Temperature, Industrial Applications of Humidification and Dehumidification, Psychrometric Charts(Reading and Interpretation), Properties of Air-Water Vapor Mixtures, Dry Bulb, Wet Bulb, and Dew Point and Adiabatic Saturation Temperatures, Enthalpy and Specific Volume of Moist Air, Types and design of Humidification Equipment: Spray Chambers, Packed Towers, and Air Washers, Types and Design of Dehumidifiers and Cooling Towers.	**8**
IV	**Adsorption:** Principle and types of adsorption, nature of adsorbents, adsorption equilibria – single gases and vapours, adsorption hysteresis, effect of temperature, heat of adsorption, adsorption of solute from dilute solution, single stage and multistage crosscurrent operation using Freundlich and Langmuir isotherm, multistage counter-current operation using Freundlich equation., stage-wise and continuous contact operations, unsteady state: Fixed-bed adsorption	**8**
V	**Crystallization:** Definition and Importance& Applications, Classification Crystallization Processes: Cooling, Evaporative, Reactive, and Precipitative Crystallization, Crystallization Equipments, Thermodynamics and Kinetics, Thermodynamics of Supersaturation: Solubility Curves and Metastable Zones,, Homogeneous vs. Heterogeneous Nucleation, Nucleation Rate, Crystal Growth: Mechanisms and Rate Equations, Factors Influencing Crystal	**8**

	Growth and Size Distribution, Types of Crystallizers: Batch, Continuous, MSMPR (Mixed-Suspension Mixed-Product Removal), DTB (Draft Tube Baffle)	
		40

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B.TECH III YEAR VI SEMESTER CHEMICAL ENGINEERING

SUBJE CT CODE: BCH 602	**COURSE TITLE: TRANSPORT PHENOMENA**
EXAM DURATION: 3 HOURS	**SEMESTER: VI (EVEN)**
L: T: P :: 3 : 1 : 0 CREDITS: 4	**PRE-REQUISITES: CEFM, HTO, MTO**

OBJECTIVE: The objective of subject is that student gain a fundamental understanding of momentum, heat, and mass transfers at the microscopic and macroscopic levels.

COURSE OUTCOMES:

After successful completion of the course, the students will be able to:

1. Understand and apply the shell balance for fluid flow problems
2. Apply the Navior-stokes equation for fluid flow problems
3. Formulate problems of heat transfer in laminar flow
4. Apply energy equation for various conditions
5. Formulate the problems of mass transfer in laminar flow
6. Understand the concept of boundary flow

CO-PO Mapping (1 to 3 scale)

CO/PO	**PO-1**	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**	**PO-7**	**PO-8**	**PO-9**	**PO-10**	**PO-11**	**PO-12**
CO-1	3	2	2	2	1	-	-	-	-	-	-	3
CO-2	3	3	2	2	1	-	1	-	-	-	-	2
CO-3	3	2	2	2	-	-	1	-	-	-	-	1
CO-4	3	2	2	3	1	1	-	-	-	-	-	1
CO-5	3	3	3	3	1	1	-	-	-	-	-	2
CO-6	3	3	3	3	-	1	-	-	-	-	-	1
Avg	**3.0**	**2.5**	**2.3**	**2.5**	**1.0**	**1.0**	**1.0**	**-**	**-**	**-**	**-**	**1.7**

REFERENCE BOOKS

S.NO	**NAME OF AUTHORS/BOOKS /PUBLISHERS**	**YEAR OF PUBLICATION/ REPRINT**
1	Bird, R. B., Stewart, W. E. and Lightfoot, E. W., Transport Phenomena, 2nd Edition, John Wiley & Sons, London (Paperback)	2007
2	Brodkey, R. S., Hershey, H. C., Transport Phenomena, Bertrams Print On Demand, Swindon, U.K., ISBN: 978-0-972-66359-5.	2003
3	Welty, J. R., Wilson, R. W., and Wicks, C. W., Gregory L. R., Fundamentals of Momentum Heat and Mass Transfer, 6th Edition. John Wiley& Sons, New Jersey/	2020

4	Chistie John Geankoplis, Transport Processes and Separation Process Principles, 5th Edition	2020

COURSE DETAILS:		
UNITS	**CONTENTS**	**LECTURE HOURS**
I	Basic Tensor Operations, Fundamental concepts in momentum transfer, shell balance, governing equations and relevant boundary conditions, Formulation and solution of momentum transfer in laminar flow	**6**
II	Navier-Stokes equation and its applications, solutions of momentum transfer problems in different geometries, Formulation and solution of heat transfer in laminar flow	**12**
III	Development and use of energy equation, Transient conduction - lumped capacitance, analytical solutions and other methods.	**6**
IV	Formulation and solution of mass transfer in laminar flow. Development and use of species balance equation	**10**
V	Introduction to convective flow, natural convection, relevant examples from heat and mass transfer	**6**
	Total	**40**

SUBJE CT CODE: BCH 603	COURSE TITLE: CHEMICAL TECHNOLOGY
EXAM DURATION: 3 HOURS	SEMESTER: VI (EVEN)
L: T: P :: 3 : 1 : 0 CREDITS: 4	PRE REQUISITES: NIL

OBJECTIVES:

1. To provide knowledge of the various processes involved in chemical industries for the production ofinorganic and organic chemicals.
2. To disseminate knowledge of the Production and Consumption Pattern, Manufacturing process and flowsheet and enable to keep up the productivity while maintaining all safety norms stipulated, during their job.
3. To make students aware of the latest technological developments and Engineering problems.

COURSE OUTCOMES:

After successful completion of the course the students will be able to:

1. Understand the principles and symbols used in process flow diagrams and their application in various industrial processes.
2. Explain the production methods and applications of key chemicals and by-products in the chlor-alkali and sulphur industries.
3. Describe the processes involved in the production of cement, fuel gases, and industrial gases, including producer gas, syn gas, and nitrogen.
4. Analyze the manufacturing processes of phosphorus, phosphoric acid, and nitrogenous fertilizers, and evaluate their role in agriculture.
5. Discuss the chemical processes for producing bio-fertilizers, pesticides, and the technology behind soap, detergents, and other oil-based products.
6. Examine the refining and secondary processing of crude oil, the manufacturing processes in the petrochemical industry, and the production of various polymers and synthetic fibers.

CO-PO Mapping (1 to 3 scale)

CO/PO	PO -1	PO -2	PO -3	PO -4	PO -5	PO -6	PO -7	PO -8	PO -9	PO -10	PO -11	PO -12
CO-1	3	2	2	1	1	2	2	-	-	-	-	1
CO-2	3	3	3	2	2	2	2	1	1	1	-	2
CO-3	3	3	3	2	2	3	3	2	2	1	-	2
CO-4	2	3	3	3	2	3	3	2	1	1	1	2
CO-5	2	3	3	3	2	3	3	2	2	1	1	2
CO-6	3	3	3	3	2	3	3	1	1	2	2	2

TEXTBOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION / REPRINT
1	Dryden's Outlines of Chemical Technology by M.G. Rao, Third edition, East West Press	2018

2	Shreve's Chemical Process Industries by G.T. Austin, Fifth edition, Tata McGraw Hill	2012
REFERENCE BOOKS/ONLINE SOURCE		
1	Moulijn J.K; Makkee M. and van Diepen A;"Chemical Process Technology", Second edition, Wiley	2013
2	Riegel's Hand Book of Industrial Chemistry by James A Kent, tenth edition, CBS publishers & distributors	2013
3	O P Gupta, "Chemical Process Technology", Khanna Publishing House.	2018
4	NPTEL Course/Notes	

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	Introduction to process equipments and major process symbols, Preparation of process flow diagrams Chlor-Alkali Industries: Production of common salt, caustic soda, chlorine, hydrochloric acid, and soda ash, Sulphur Industries: Origin and extraction of sulfur, production routes of sulphuric acid and oleum. Cement, Fuel, and Industrial Gases: Technology options of producing producer gas, syn gas, pyrogas, nitrogen, oxygen and carbon dioxide.	**10**
II	Phosphorous Industries : Manufacturing of phosphorus, phosphoric acid and phosphatic fertilizers. Nitrogen Industries : Manufacturing of ammonia, nitric acid, nitrogenous and mixed fertilizers, Bio-fertilizers, Pesticides: Manufacturing process of insecticides, fungicides, and herbicides.	**10**
III	Oil fats and waxes industry: properties of oils and fats, Saturated and unsaturated fatty acids, hydrogenation of edible oils, Soap & detergent, Pulp & paper, wood composites	**6**
IV	Petroleum Industry : Origin, occurrence and characteristics of crude oil, crude oil distillation and secondary processing. Petrochemical Industry: Manufacturing processes of formaldehyde, acetaldehyde, acetic acid, acetic anhydride, maleic anhydride, nitrobenzene, ethylene oxide, ethylene glycol, ethanolamine, cumene, ethyl benzene, BTX	**10**
V	Polymer and Synthetic Fibre : Introduction to polymerization, the manufacturing process of polymers commodity polymers, rayon, polyester, polyamide, acrylic fibre and nylons.	**4**
	TOTAL	**40**

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B.TECH III YEAR VI SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 651	**COURSE TITLE: CHEMICAL TECHNOLOGY LAB**
EXAM DURATION: 2 HOURS	**SEMESTER: VI (EVEN)**
L: T: P :: 0 : 0 : 2 CREDIT: 1	**PRE-REQUISITE: NIL**

OBJECTIVES:

1. To introduce the fundamental principles of chemical technology lab experiments.
2. To make them correlate theory and practical processes in industry, through experimentation.
3. To enhance their practical knowledge and thus their employability.

COURSE OUTCOMES:

After successful completion of this course, the students will be able to:

1. Understand the processes involved in the preparation of Turkey Red Oil and its applications in the leather industry.
2. Analyze the formulation and production techniques for dry/oil-bound distemper and cement paint.
3. Explore the manufacturing processes of liquid soap, transparent soaps, and detergent powder.
4. Study the preparation of alkyd resin and its applications in coatings and paints.
5. Gain insights into the processes of oil extraction, splitting, and margarine production.
6. Learn the techniques for preparing bio-fertilizers and sanitisers, focusing on their industrial applications.

CO-PO Mapping (1 to 3 scale)

CO/PO	PO -1	PO -2	PO -3	PO -4	PO -5	PO -6	PO -7	PO -8	PO -9	PO-10	PO-11	PO-12
CO-1	3	2	2	1	1	2	2	-	1	-	-	2
CO-2	3	2	3	2	2	-	1	-	1	-	-	2
CO-3	3	2	3	2	2	-	-	-	1	-	-	2
CO-4	3	2	3	2	2	1	-	-	1	-	-	2
CO-5	3	2	3	2	2	-	1	-	1	-	-	2
CO-6	3	2	3	2	2	-	1	-	1	-	-	2
Avg	**3.0**	**2.0**	**2.8**	**1.8**	**1.8**	**1.5**	**1.3**	**-**	**1**	**-**	**-**	**2**

REFERENCE BOOKS

S.NO	**NAME OF AUTHORS/BOOKS /PUBLISHERS**	**YEAR OF PUBLICATION/ REPRINT**
1	Dryden, C. E. "Outlines of Chemical Technology" (Edited and Revised by M. Gopala Rao and M. Sittig)East West Press. Pvt. Ltd, New Delhi, 3rd Edition	2007

2	Austin G. T. Shreve's Chemical Process Industries", 5th Edition, McGraw Hill	1984
3	"Chemtech" Volume I - IV, 4th Edition, Chemical Engineering Education Development Centre, I.I.T., Madras.	2017
4	O P Gupta, "Chemical Process Technology", 8th Edition, Khanna Publishing House.	2023

COURSE DETAILS:

LIST OF EXPERIMENTS:

Preparation and Quality evaluation of any ten experiments of the following:

1. Preparation of Turkey Red Oil for leather industry from castor oil.
2. Preparation of dry/oil bound distemper.
3. Preparation of cement Paint.
4. Preparation of Liquid soap.
5. Preparation of alkyd resin.
6. Preparation of Transparent Soaps.
7. Preparation of Detergent Powder.
8. Preparation of Margarine.
9. Soxhlet Extraction of oil bearing materials.
10. Splitting of Oils for fatty acid preparation.
11. Preparation of bio-fertilizer.
12. Preparation of Sanitizer.

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B. TECH III YEAR VI SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 652	**COURSE TITLE: MASS TRANSFER-II LAB**
EXAM DURATION: 2 HOURS	**SEMESTER: VI (EVEN)**
L: T: P :: 0 : 0 : 2 CREDIT: 1	**PRE-REQUISITE: NIL**

OBJECTIVES:

1. To impart knowledge of the basic fundamental principles of mass transfer by performing different experiments
2. To make them correlate theory and practical process by experimentation.

LAB OUTCOMES:

On successful completion of the lab, the student will be able to:

1. Analyze the data on Analyze vapor-liquid equilibrium and Boiling point diagram.
2. Discuss the performance of the distillation column.
3. Explain the adsorption kinetics and isotherm at the Solid-Liquid interface.
4. Understand the separation process by Liquid-Liquid Extraction and solid-liquid extraction.
5. Discuss the crystallization process.
6. Discuss the humidification process.

REFERENCE BOOKS:

S. NO.	NAME OF AUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	Robert. E. Treybal. ―Mass Transfer Operations II, 3rd edition, Mc Graw Hill, NY,	1980
2.	McCabe and J.M.Smith. ―Unit Operations in Chemical Engineering II, 7th edition, McGraw Hill	2005
3.	Coulson and Richardson ―Heat and Mass Transfer: Fundamentals and Applications II, Vol I-B, 7edition	2016
4.	J.D. Seader & Henley E. J., "Separation Process Principles II,2nd edition, Wiley India Pvt. Ltd,	2006
5.	Geankoplis, C.J. ―Transport Processes and Unit Operations II, 3e, Prentice Hall (I),	1978

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B. TECH III YEAR VI SEMESTER CHEMICAL ENGINEERING

COURSE DETAILS:

S. No.	LIST OF EXPERIMENTS
1	To study vapor-liquid equilibrium and prepare Boiling point diagram for a binary liquid mixture.
2	To study the performance of Bubble cap distillation column.
3	To verify Rayleigh equation for differential distillation of binary system.
4	To study the performance of sieve plate distillation unit.
5	To determine Overall efficiency for a three-stage counter-current and cross current system.
6	To determine the ternary curve for the system acetic acid-water-carbon tetrachloride.
7	To study the solid –liquid extraction system- Soxhlet's experiment.
8	To study the operation on extraction of oil from seed.
9	To determine the adsorption kinetics and isotherm at solid-liquid interface.
10	To obtain the breakthrough curve for the given adsorption system.

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B.TECH III YEAR VI SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 653	**COURSE TITLE: TECHNICAL PRESENTATION**
EXAM DURATION: 2 HOURS	**SEMESTER: VI (EVEN)**
L: T: P :: 0 : 0 : 2 CREDIT: 1	**PREREQUISITE: NIL**

OBJECTIVES: The purpose of this course is to prepare our students for:

1. To provide them with better learning and understanding through presentations.
2. To develop better communication skills and confidence.
3. To provide a platform for sound discussions of technical & challenging areas.

COURSE OUTCOMES

On successful completion of the lab, the student will be able to:

1. Improve their communication skills.
2. How to write refined reports of any technical topics.
3. To learn new challenging areas of their domain.
4. Knowledge of the application of Artificial Intelligence in Chemical Engineering.
5. Knowledge of automation through ERP module training.
6. Express and write thoughts in own language

CO-PO Mapping (1 to 3 scale)

CO/PO	PO -1	PO -2	PO -3	PO -4	PO -5	PO -6	PO -7	PO -8	PO-9	PO-10	PO-11	PO-12
CO-1	1	1	1	1	1	1	-	3	1	2	3	3
CO-2	2	1	1	-	1	-	-	3	1	2	3	3
CO-3	1	2	1	1	2	2	-	3	2	2	2	3
CO-4	1	1	1	1	1	1	-	2	2	2	2	2
CO-5	-	-	-	2	2	1	-	2	1	2	3	2
CO-6	-	-	-	1	1	1	-	3	1	3	2	2
Avg	**1.3**	**1.3**	**1.0**	**1.2**	**1.3**	**1.2**	**-**	**2.7**	**1.3**	**2.2**	**2.5**	**2.5**

REFERENCE BOOKS:

S. NO.	NAME OF AUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	Kumar, Sanjay & Pushp Lata, "Communication Skills", Oxford University Press. 2nd Edition	2017
2.	Quirk & Randolph, "A University Grammar of English", Pearson. 5th Edition	2022
3.	Rutherford, Andrea J., "Basic Communication Skills for Technology", Pearson. 2nd Edition	2016
4.	Rizvi, M Ashraf, "Effective Technical Communication", McGraw Hill. 3rd Edition	2019

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B.TECH III YEAR VI SEMESTER CHEMICAL ENGINEERING

Course Content :
Preparation of a PowerPoint presentation on any of the latest awareness topics related to Chemical Engineering, entrepreneurship, start-ups, environment sustainability, etc For this purpose, the following references can also be used: Vikasit Bharat-2047 Net Zero Industry Act Winning Leap-2034 Indian Knowledge Systems Start-up India scheme Technology Readiness Level Production Linked Incentive Scheme Sustainability Development Goals Circular economy Other technical innovation topics can also be taken which generates an Out of the box thinking.

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

B.TECH III YEAR VI SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 061	**COURSE TITTLE: FUNDAMENTALS OF POLYMER SCIENCE AND ENGINEERING**
EXAM DURATION: 3 HOURS	**SEMESTER: VI (EVEN)**
L: T: P :: 3 : 0 : 0 CREDITS: 3	**PREREQUISITE: NIL**

OBJECTIVES:

1. To provide a broad and fundamental knowledge of polymers and their chemical, physical, and mechanical behavior.
2. To emphasize polymer synthesis, reaction engineering, and various processing techniques like moulding and extrusion and learn the various mechanisms of polymerization, and choose suitable techniques for polymer synthesis.
3. To equip the students with the knowledge necessary for deciding which characterization technique(s) would be appropriate for determining properties of interest.

COURSE OUTCOMES:

On successful completion of the course, the student will be able to:

1. Understand the fundamental principles of polymer chemistry, including polymerization reactions, functionalities, and different types of polymerizations such as free radical and ionic.
2. Analyze the kinetics and mechanisms of radical, chain, and ionic polymerization, and explore the complexities of copolymerization and stereospecific polymerization.
3. Evaluate techniques for estimating the molecular weight of polymers and apply this knowledge to different polymerization processes including bulk, solution, emulsion, and suspension polymerization.
4. Identify and describe various types of polymers, including engineering polymers, thermoplastics, thermosets, and elastomers, and their applications in diverse fields.
5. Examine methods for polymer characterization, including key techniques and instruments, and understand the role of BIS standards in polymer quality and performance.
6. Explore polymer processing techniques and manufacturing methods, including extrusion, injection molding, and recycling, and analyze the use of additives, blending, and compatibilization in polymer products and composites.

CO-PO Mapping (1 to 3 scale)

CO/PO	**PO -1**	**PO -2**	**PO -3**	**PO -4**	**PO -5**	**PO -6**	**PO -7**	**PO -8**	**PO -9**	**PO -10**	**PO -11**	**PO -12**
CO-1	3	2	2	1	2	1	2	1	1	1	1	2
CO-2	3	3	2	2	3	1	1	1	1	1	1	2
CO-3	3	2	3	2	3	2	2	1	1	1	2	3
CO-4	3	2	2	2	3	1	2	1	1	1	1	2
CO-5	3	3	2	2	3	2	1	1	1	1	1	3
CO-6	3	2	2	3	3	1	2	1	1	1	2	3
Avg	**3.0**	**2.3**	**2.2**	**2.0**	**2.8**	**1.3**	**1.7**	**1.0**	**1.0**	**1.0**	**1.3**	**2.5**

REFERENCE BOOKS:

S. NO.	NAME OF AUTHORS/BOOKS/PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	F.W. Billmeyer, "Text Book of Polymer Science", 3rd edition ., Wiley Inter-Science	1984
2.	F. Rodriguez, "Principles of polymer systems",4th edition, Taylor, and Francis, Washington	2003
3.	Herman F. Mark (Editor) "Encyclopaedia of Polymers Science and Technology", John Wiley-Inter Science. 4th edition .	2005
4.	J. R. Fried, "Polymer Science and Technology", Prentice-Hall, Inc .1st edition 1973	1973
5.	A. Kumar, "Fundamentals of Polymer Engineering", 2nd editon.	2002
6.	A. Rudin , "Elements of Polymer Science and Engineering",2nd editon. Academic Press, San Diego London Boston New York Sydney Tokyo Toronto.	1990

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	Fundamental of Polymer Chemistry Chemistry of Polymerisation Reaction: Functionality, polymerization reactions, polycondensation, addition free radical and chain polymerization, copolymerization, block and graft polymerizations, stereo specific polymerization. Kinetics of radial, chain and ionic polymerization and co-polymerisation systems. Molecular Weight Estimation: Average molecular weight Polymerisation Processes: Bulk, solution, emulsion and suspension polymerization. Thermoplastic composites, fibre reinforcement fillers, surface treatment, reinforced thermoset composites-resins, fibers additives, fabrication methods.	**8**
II	Types of polymers; Blends; Composites; Materials and applications of polymers Engineering Polymers: Types of engineering polymers; Thermoplastics; Thermosets; Elastomers; Polyolefins; Polyethylene; Polypropylene, High Performance Polymers, Polyesters; Polycarbonate; Polyamide; Polyetherimides; Poly-amide-imide; High temperature Resistant polymers	**9**
III	Characterization Of Polymers: Some important techniques and instruments used for characterization of polymers, Introduction of BIS Standards for polymers Polymerization engineering; Polymer Processing; Additives; Polymer products; Processing; Introduction to rheology and various polymer processing techniques and equipments	**8**

IV	Manufacturing Of Polymers: Polyethylene, polypropylene, polyvinylchloride and copolymers, polystyrene, polyamides, polyesters, Acrylics, Phenol-formaldehyde, Melamine-formaldehyde, Polyurethane, Epoxides, Rubbers and elastomers Polymer Mixtures: Polymer compatibilization, Thermodynamic theories for polymers solutions, solubility parameter, solvents and plasticizers, Polymer blending, reinforced plastics and elastomers, additives for polymers, polymer composites	**9**
V	Plastics: extrusion, injection molding, blowmolding, compression, and transfer molding, Recycling of polymers Applications of polymers in commodity products	**6**
		40

DR. A.P. J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

B.TECH. III YEAR VI SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 062	**COURSE TITLE: SUSTAINABILITY OF ENVIRONMENT**
EXAM DURATION: 3 HOURS	**SEMESTER: VI (EVEN)**
L: T: P :: 3 : 0 : 0 CREDITS: 3	**PREREQUISITE: NIL**

OBJECTIVES:

1. To create awareness in every engineering graduate about the significance of sustainability of environment.
2. To teach about the effect of technology on the environment and ecological balance.
3. To make students sensitive to the sustainable utilization of natural resources.

COURSE OUTCOMES:

On successful completion of the course, the student will be able to:

1. Understand the interaction between humans and the environment, including key ecological concepts, biodiversity, and pollution from chemical industries.
2. Apply environmental laws, standards, and regulations for air, noise, and effluent management as set by control boards like CPCB and UPPCB.
3. Implement resource conservation strategies using the 5R principles, alternative materials, and waste recovery methods for energy and water conservation.
4. Analyze air and water pollution sources, parameters, and control measures, including air pollution control equipment and wastewater treatment processes.
5. Manage solid waste through effective practices for hazardous and non-hazardous waste, addressing plastic and e-waste challenges, and adhering to waste management laws.
6. Evaluate sustainable development practices, including cleaner technologies, life cycle assessment, and case studies on energy conservation and renewable energy technologies.

CO-PO Mapping (1 to 3 scale)

CO/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	2	2	2	1	3	3	1	1	1	1	2
CO-2	3	3	2	2	2	3	3	1	1	1	1	2
CO-3	3	3	3	2	2	3	3	1	1	2	1	3
CO-4	3	3	3	3	3	3	3	1	1	2	1	3
CO-5	3	3	2	2	2	3	3	1	1	1	2	3
CO-6	3	2	2	2	2	3	3	1	1	2	2	3
Avg	**3.0**	**2.7**	**2.3**	**2.2**	**2.0**	**3.0**	**3.0**	**1.0**	**1.0**	**1.5**	**1.3**	**2.7**

REFERENCE BOOKS:

S. NO.	NAME OF AUTHORS/ BOOKS/ PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	Howard S. Peavy, D. R. Rowe & C. Tchobonoglous "Environmental Engineering", McGraw Hill 5th edition	2023
2.	Metcalf & Eddy, "Wastewater Engineering Treatment, Disposal & Reuse", Tata McGraw Hill. 5th Edition	2014
3.	Pandey G. N. and Carney G. C., "Environmental Engineering ". Tata McGraw Hill. 2nd edition	2018
4.	Kreith F. and Tchobanoglous G., "Handbook of Solid Waste Management", 2nd edition, Mc Graw Hill	2000
5.	Werner Strauss, 'Air Pollution Control: Measuring and monitoring air pollutant' Wiley. 2nd edition	2018
6	Pichtel J., "Waste Management Practices: Municipal, Hazardous and Industrial", CRC Press 3rd edition.	2019

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	**Environment and Biodiversity:** Definition, scope and importance of environment - need for public awareness., Ecology & Environment, components of the biosphere, biodiversity, Food chain, Environmental pollution from chemical process industries, Role of chemical engineering in sustainability, Carbon footprint and greenhouse gases, characterization of emission and effluents, environmental Laws,	**8**
II	**Resource Conservation:** Process modification, Concept of 5R (Refuse, Reduce, Reuse, Repurpose, Recycle) alternative raw material, recovery of by co-product from industrial emission effluents, recycle and reuse of waste, energy recovery and waste utilization, Water use minimization, Emerging technologies and future trends for sustainable resource conservations	**6**
III	Environmental Pollution: Air, Water, Soil, Noise Pollution, Various pollution parameters like BOD, COD, pH, total suspended solids, turbidity, total solids, standards for ambient air, noise emission, and various effluents, Pollution control technologies, Wastewater treatment like primary, secondary and tertiary, Acts and rules by control boards (CPCB and UPPCB), the concept of pollution prevention for sustainable development	**8**

IV	Solid Waste Management: Solid waste generation, sources and characteristics of Industrial, municipal, hazardous and organic and plastic and E-waste. Collection, handling, storage, and disposal of various solid wastes, waste management laws, and guidelines. Non-hazardous industrial wastes-treatment, disposal, utilization, and management.	**8**
V	**Sustainability Practices:** Sustainable Development Goals, Environmental impact monitoring, Eco-friendly packaging, Carbon footprint reduction, Waste minimization, Circular economy principles, Material Life cycle assessment. Sustainable habitat: Green buildings, green materials, sustainable transports. Sustainable energy: Non - conventional Sources. Green Engineering, Sustainable urbanization Case studies on techno-economics sustainable technologies for making non-renewable energy sources available over longer periods. Technical Case studies on the Ministry of Environment, Forest and Climate Change, UPPCB and CPCB, and other sites.	**10**
	TOTAL	**40**

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B.TECH III YEAR VI SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 063	**COURSE TITLE: COLLOID SURFACE & INTERFACIAL PHENOMENA**
EXAM DURATION: 3 HOURS	**SEMESTER: VI (EVEN)**
L: T: P :: 3 : 0 : 0 CREDITS: 3	**PRE-REQUISITES: NIL**

OBJECTIVES:

1. To introduce students to the fundamentals of colloidal and interfacial phenomena
2. Exposing them to a broad selection of topics, including colloidal suspensions, Intermolecular, nanoscaleand interfacial forces systems
3. To provide knowledge about the importance of the surface properties.

COURSE OUTCOMES :

On successful completion of the course, the student will be able to:

1. Understand the classification and stabilization mechanisms of colloidal systems using surfactants.
2. Analyze surfactant behavior in micellization, emulsification, and micellar catalysis.
3. Explore fundamental aspects of colloidal suspensions, including surface tension and surfactant adsorption.
4. Evaluate intermolecular and interfacial forces in organic, polymeric, and biological systems.
5. Apply mesoscale thermodynamics to soft matter phenomena, such as adhesion and wetting.
6. Investigate nanofluidics and advanced interfaces, focusing on thin film stability and functional coatings.

CO-PO Mapping (1 to 3 scale)

CO/ PO	**PO -1**	**PO -2**	**PO -3**	**PO -4**	**PO -5**	**PO -6**	**PO -7**	**PO -8**	**PO -9**	**PO -10**	**PO -11**	**PO -12**
CO-1	3	3	2	2	2	2	2	-	-	-	-	2
CO-2	3	3	2	2	3	2	2	-	-	-	-	2
CO-3	3	3	2	2	3	3	2	-	-	-	-	2
CO-4	3	3	2	2	3	3	3	-	-	-	-	2
CO-5	3	3	2	2	3	3	3	-	-	-	-	2
CO-6	3	3	2	2	3	3	3	-	-	-	-	2
Avg	**3.0**	**3.0**	**2.0**	**2.0**	**2.8**	**2.7**	**2.5**	-	-	-	-	**2**

REFERENCE BOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	Surfactants and interfacial phenomena, Milton J. Rosen, 5th edition	2012
2.	Principles of Colloid and Surface Chemistry, Paul C. Hiemenz, Marcel DEker,2nd edition	2007
3.	Physical Chemistry of Surfaces, Arthur W. Adamson, 6th Edition Wiley.	1997
4.	Foundations of Colloid Science, Robert J. Hunter, 2nd edition (2001) Oxford University Press	2001
5.	Colloidal Dispersions, W. B. Russel, D. A. Saville, and W, R. Schowalter,Cambridge University Press . 3rd Edition 2012	2012
6.	Intermolecular and Surface forces, Jacon N. Israelachvili, Academic Press, 3rd edition	2011
7.	Interfacial Forces in Aqueous Media, Carel J. van Oss, Marcel Dekker or Taylor Francis. 2nd edition	2006

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	Introduction of colloidal systems and their classification, origin of charge on colloidal particles, stabilization of different colloidal system by surfactants, solubilisation, emulsions, microemulsions, HLB, micelle formation by surfactants, thermodynamic parameters of micellization, emulsification by surfactants, micellar catalysis, synergism in mixture of two surfactants.	**8**
II	Fundamental aspects of colloidal suspensions, surface tension, wetting, surfactant adsorption, self-assembly, and inter particle interactions, as well as the importance of these phenomena to consumer, industrial, and biomedical applications.	**6**
III	Intermolecular, nanoscale and interfacial forces in organic, polymeric, biological and aqueous systems. Van der waals, electrostatic double layer, acid-base interactions including hydrophobic attraction and hydration pressure.	**8**
IV	Mesoscale thermodynamics and Mesoscale phenomena in soft matter & applications Gibb's treatment of interfaces, concept of excess concentration, variation of interfacial tension with surface concentration, Adhesion, wetting, nucleation, flotation, patterning ofsoft material by self organization and other techniques.	**10**
V	Nanofluidics and Advanced & Functional Interfaces: Stability of thin (< 100 nm) film, self-organization in confined systems, mesoscale patterning. Super hydrophobicity, functions coatings, structural colours, nano-adhesives, nanocomposites.	**8**
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DR. A.P. J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH. III YEAR VI SEMESTER CHEMICAL ENGINEERING

SUBJECT CODE: BCH 064	**COURSE TITLE: ENVIRONMENTAL IMPACTASSESMENT**
EXAM DURATION: 3 HOURS	**SEMESTER: VI (EVEN)**
L: T: P :: 3 : 0 : 0 CREDITS: 3	**PRE-REQUISITE: NIL**

OBJECTIVES:

1. To give knowledge the different steps within environmental impact assessment.
2. To provide a broad understanding of environmental auditing and environmental impact assessment.
3. To impart knowledge about the increasing importance of corporate social responsibility.

COURSE OUTCOMES:

On completion of this course, the students will be able to:

1. Comprehend the basics of Environmental Impact Assessment (EIA) and its role in industrial operations.
2. Analyze the roles of regulatory agencies, engineering firms, and public participation in EIA planning.
3. Understand the Environmental Protection Rules of 1986 and their 1993 amendment.
4. Develop proficiency in environmental auditing methods and procedures, with a focus on major industries.
5. Explore the constitutional and legislative framework for pollution prevention and control in India.
6. Gain practical knowledge of environmental clearance processes, standards, and management plans through EIA case studies.

CO-PO Mapping (1 to 3 scale)

CO/PO	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12
CO-1	3	3	2	2	2	2	2	-	-	-	-	2
CO-2	3	3	2	2	3	2	2	-	-	-	-	2
CO-3	3	3	2	2	3	3	2	-	-	-	-	2
CO-4	3	3	2	2	3	3	3	-	-	-	-	2
CO-5	3	3	2	2	3	3	3	-	-	-	-	2
CO-6	3	3	2	2	3	3	3	-	-	-	-	2
Avg	**3.0**	**3.0**	**2.0**	**2.0**	**2.8**	**2.7**	**2.5**	**-**	**-**	**-**	**-**	**2.0**

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B.TECH III YEAR VI SEMESTER CHEMICAL ENGINEERING

REFERENCE BOOKS:

S. NO.	NAME OF AUTHORS / BOOKS / PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	A K Srivastava, Environment impact Assessment, APH Publishing	2015
2.	John Glasson, Riki Therivel & S Andrew Chadwick "Introduction to EIA"University College London Press Limited	2019
3.	Larry W Canter, "Environmental Impact Assessment", McGraw Hill Inc. , NewYork,	1996
4.	Ministry of Environment & Forests, Govt. of India 2006 EIA Notification	2006
5.	Rau G J and Wooten C.D "EIA Analysis Hand Book" Mc Graw Hill	
6.	Robert A Corbett "Standard Handbook of Environmental Engineering" McGrawHill.	1999
7.	O P Gupta, "Elements of Environmental pollution & Control" Khanna Publisïnghouse.	2018
8.	Peter Wathern, Routledge, "Environmental Impact Assessment (Theory andPractice)", (Taylor and Frances Group), London and New York.	1988 (digitized 2023)
9.	Brady, J., Ebbage, A. & Lunn, R., "Environmental Management in Organizations: The IEMA Handbook" 2nd edition. London, Earthscan.	2011

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B.TECH III YEAR VI SEMESTER CHEMICAL ENGINEERING

COURSE DETAILS:

UNITS	CONTENTS	LECTURE HOURS
I	Introduction to EIA: Concept of EIA, Significance of EIA in sustainable development, Flow diagram and key stages, Audit, Environment & Industries, Input information, Plant Operation, Environmental Management planning, Waste Streams impact on water bodies.	6
II	Environmental Impact Assessment: Planning. Activities, Methodology for Environmental Impact Assessment, Role of Environmental Engineering firm, Regulatory agencies & control boards like CPCB, Requirement of environment audit statement, Role of the Public. Study of Environmental Protection Rules 1986 and its amendment in 1993.	10
III	Environmental Audit: Introduction, Environmental information, Purpose & advantage of studies, General approach of environmental Auditing and Methods, Audit programs in India, Auditing program in major polluting Industries, Reports of the Environmental audit studies.	8
IV	Environmental Management: Preventive policy of environment, waste minimization, conservation of water and energy, use of renewable sources, pollution audit, pollution control strategy, disposal of treated effluents, Administrative & legislative arrangement for Environmental protection, Indian Standards, fiscal incentives for environmental protection.	8
V	Environmental Clearance process in India - Key Elements in 2006 EIA (Govt. of India) Notification, Guidelines of preparation of project report and its evaluation, clearancefrom the concern authorities (DOE, MOEF&CC) . Case studies of EIA	8
		40